



TRILYTICS 2026

PMAF

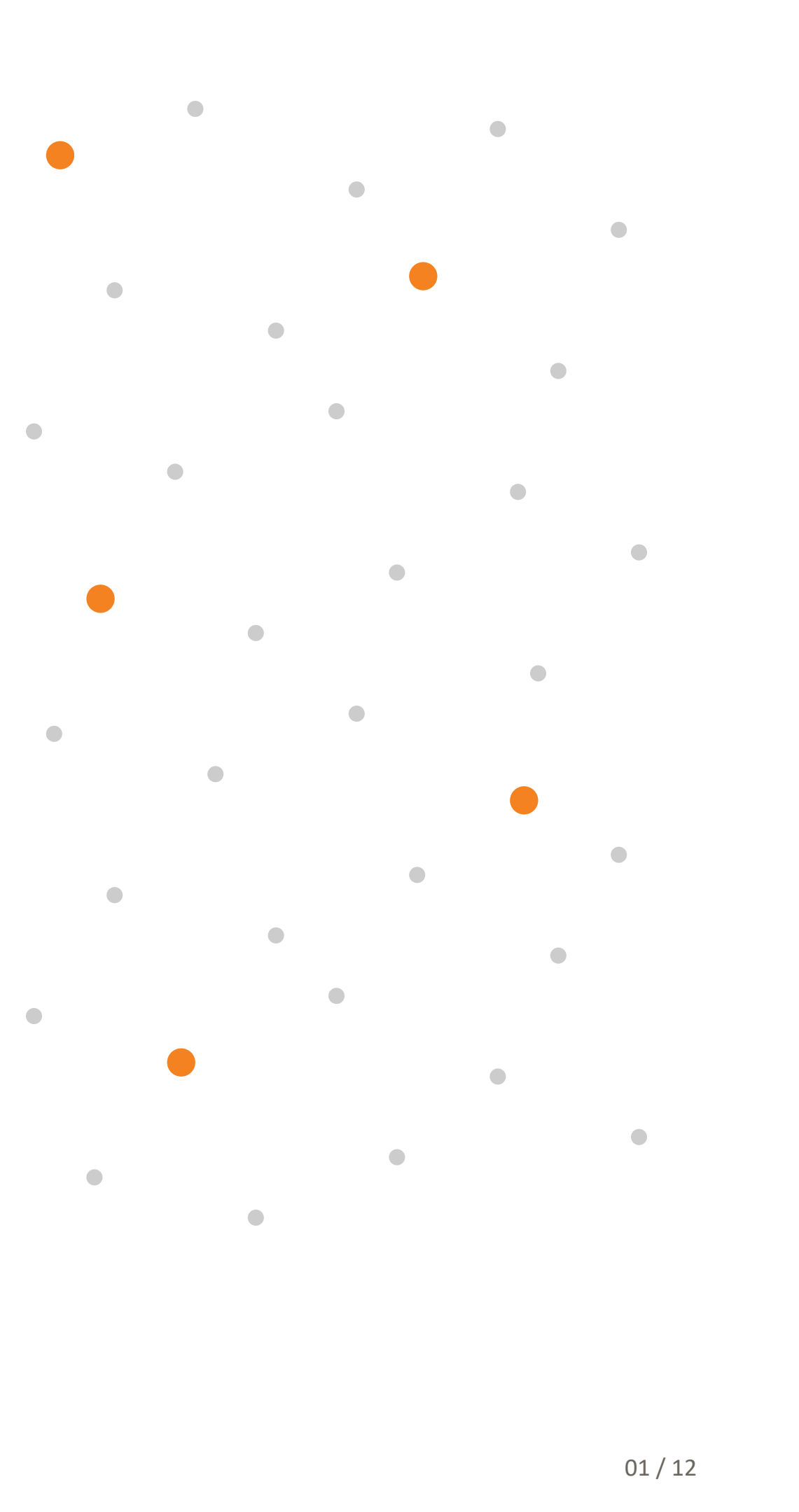
Pharmaceutical Market Attractiveness Framework

A Data-Driven District-Level Market Intelligence Framework for Sun Pharma

District-Level Opportunity Scores
Overall • Chronic • Acute

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Executive Summary



Problem

No unified framework for district-level pharmaceutical market prioritization.



PMAF Framework

Built on **851 harmonized State–District** analytical records using public datasets.



Data

53 engineered features •
6 pillars • 5 public datasets.



Outputs

Overall PMAF, Chronic PMAF and Acute PMAF with complete district rankings.



Dashboard

Interactive dashboard for exploration, comparison and territory planning.

***PMAF** translates macroeconomic, healthcare and disease-burden data into a single, comparable opportunity score.*



Business Problem: District Market Prioritization



Sun Pharma's field strategies lack a unified, data-driven district selection process.



Existing approaches rely on fragmented indicators with no therapy-specific focus.



PMAF addresses this by **integrating health and socio-economic data into one comparable score.**



Goal: Identify high-opportunity markets for chronic and acute therapies using harmonized district-level public data.

From Data to Decisions



Raw Data

Census · NFHS · RBI · SECC



PMAF Analytics Engine

Cleaning · Normalization · Scoring



Actionable Insights & Rankings

851 harmonized State–District records, ranked & scored

Data Sources & Processing



Census of India

Population, households, literacy



NFHS-5 (2019-21)

Disease burden, maternal health



Rural Health Statistics

Facility counts, infrastructure



SECC

Economic and social indicators



RBI Banking Statistics

Deposits, credit, penetration

Processing Pipeline

1

Data Collection

Assembled district-level data from 5 public sources.

2

Cleaning & Harmonization

Standardized district/state names to 851 harmonized State–District records.

3

Missing Values

Imputed using median strategy — no zero-imputation.

4

Normalization

Min-Max scaling of each feature to [0, 1]: $x' = (x - \min) / (\max - \min)$

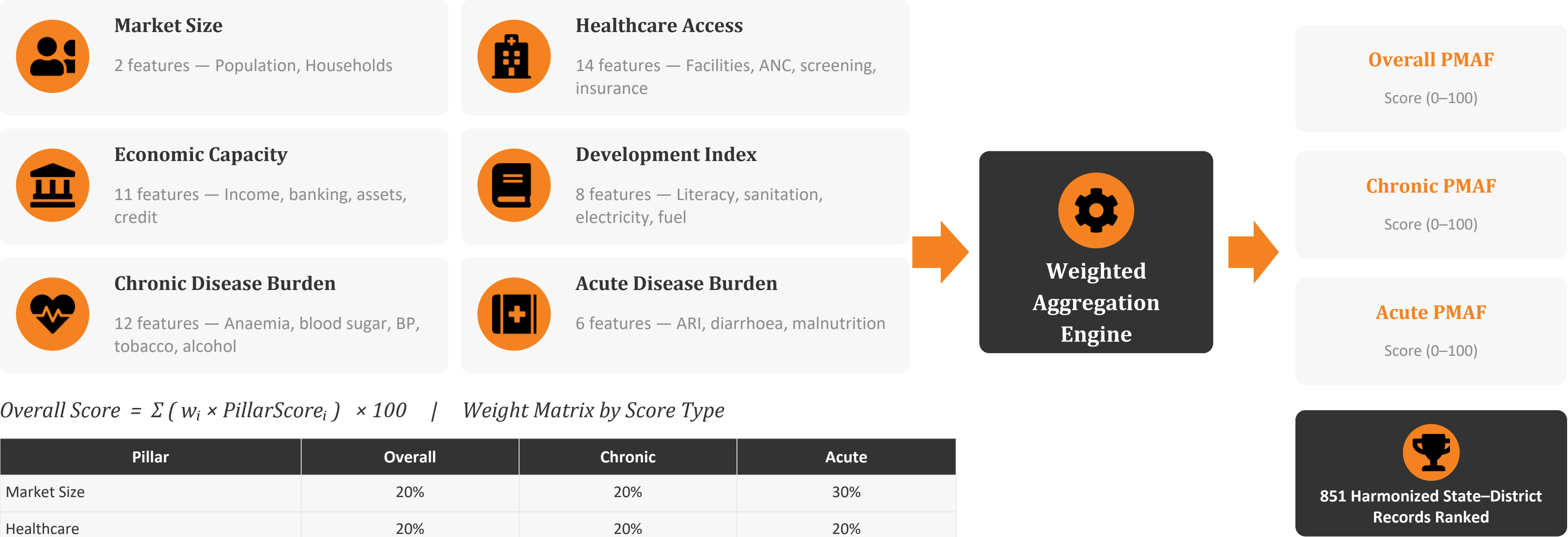
5

Reverse Scoring (Landless Manual Labour only)

Applied only to Landless Manual Labour (poverty proxy). Disease burden is **NOT** reversed — it is treated as a positive demand driver, since higher prevalence means larger pharmaceutical demand.



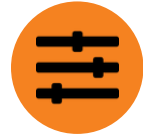
PMAF: Six Pillars to District Opportunity



$$\text{Overall Score} = \sum (w_i \times \text{PillarScore}_i) \times 100 \quad / \quad \text{Weight Matrix by Score Type}$$

Pillar	Overall	Chronic	Acute
Market Size	20%	20%	30%
Healthcare	20%	20%	20%
Economic	20%	15%	10%
Development	20%	10%	15%
Chronic Disease	10%	35%	0%
Acute Disease	10%	0%	25%

Methodology & Robustness

	Collect	Assemble 5 public datasets
	Preprocess	Clean & harmonize to 851 State–District records
	Feature Engineering	53 features across 6 pillars
	Normalize	Min-Max scale each feature to [0,1]
	Weight & Score	Apply pillar weights per score type
	Aggregate	Compute Overall / Chronic / Acute PMAF
	Rank Districts	Rank all 851 State–District records per score
	Dashboard	Publish to interactive Streamlit app

Validation & Robustness

● Explainability

Every pillar uses intuitive, publicly sourced metrics. Weights reflect relative importance validated against domain logic for chronic vs. acute markets.

● Sanity Checks

Highest-opportunity districts (Pune, Thane, Ernakulam) align with major metro and industrial regions with large, diverse populations.

● Data Integrity

Automated validation at every stage: no duplicate districts, no missing identifiers, all normalized features bounded to [0, 1].

● Scalability

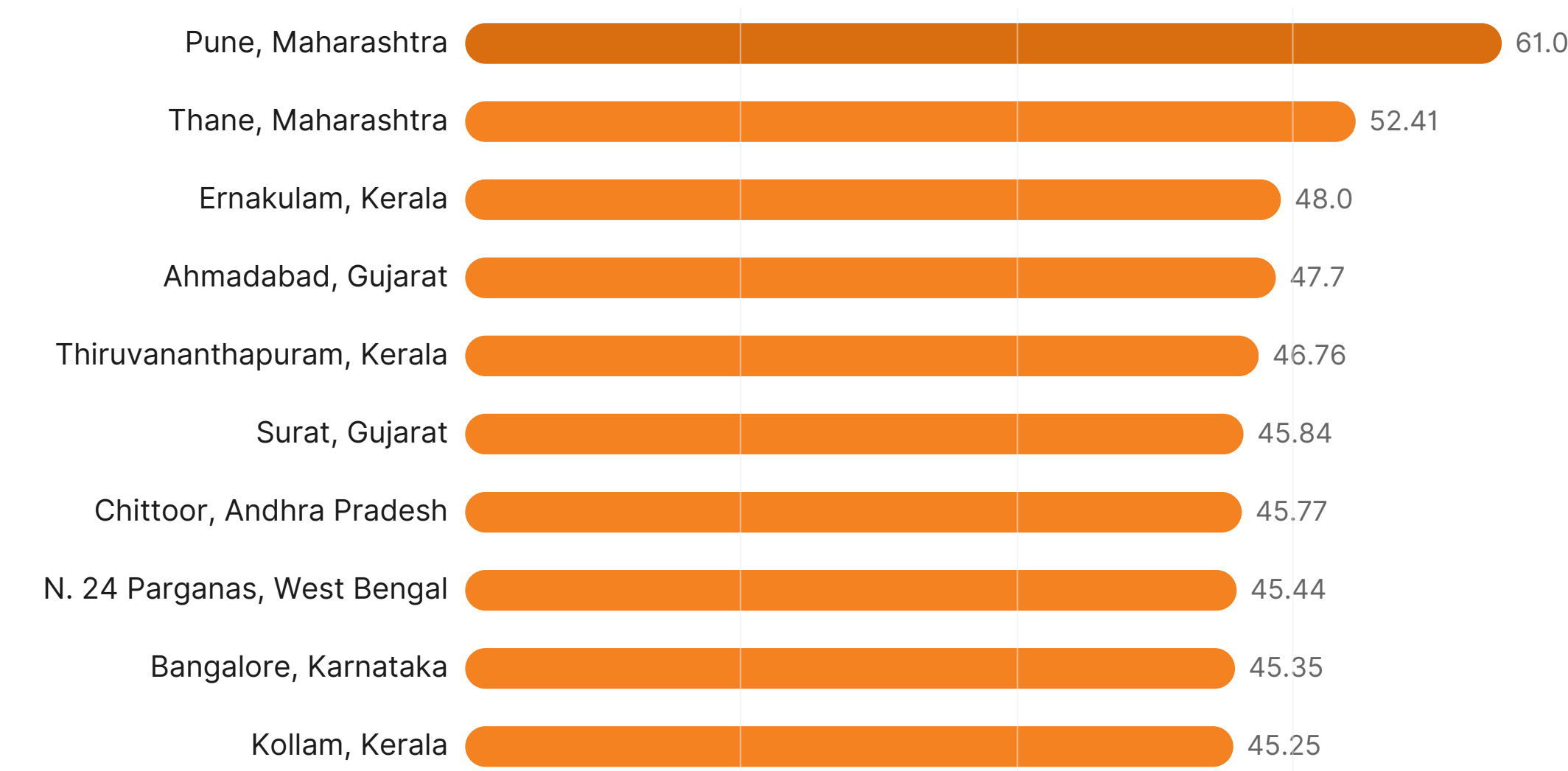
Modular pipeline (feature_engineering.py, normalize_features.py, calculate_pmaf_score.py) allows new pillars or updated data with minimal changes.

● Updatability

Data sources can be refreshed as new NFHS/Census releases become available — pipeline re-runs end-to-end.

Overall Market Opportunity: Top Districts

Top 10 Districts by Overall PMAF



61.03
Highest PMAF
Pune, Maharashtra

30.43
Average PMAF
Across 851 harmonized State–District records

0.2%
Records with PMAF > 50
2 records

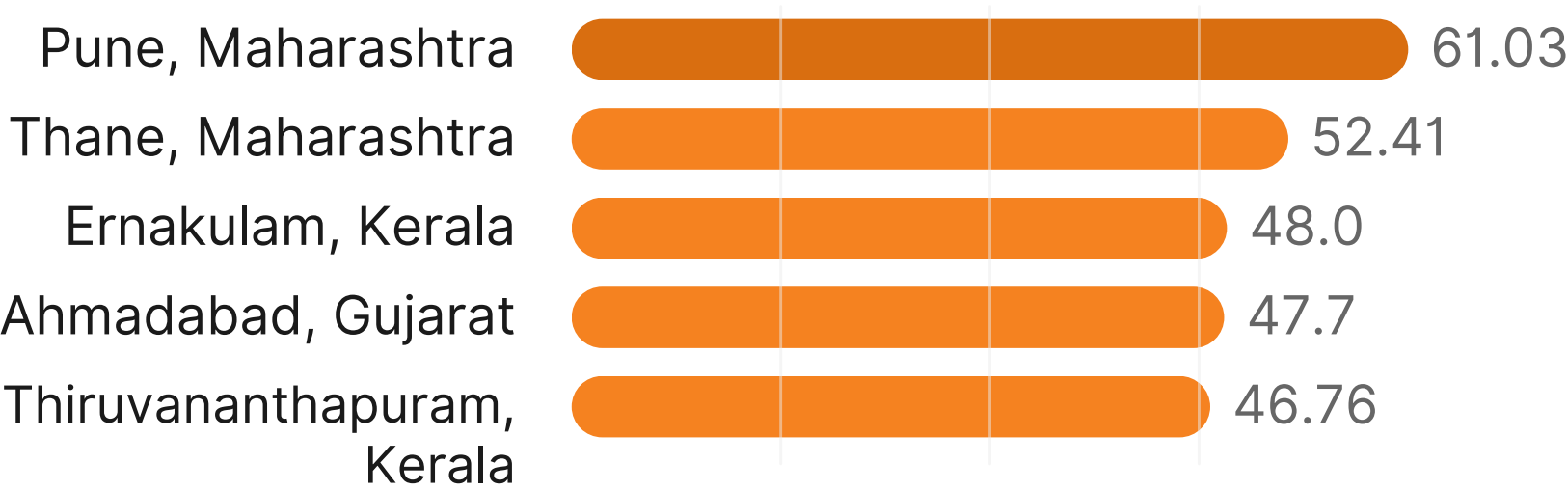
851
Harmonized State–District Records
Scored & Ranked

Key Insights

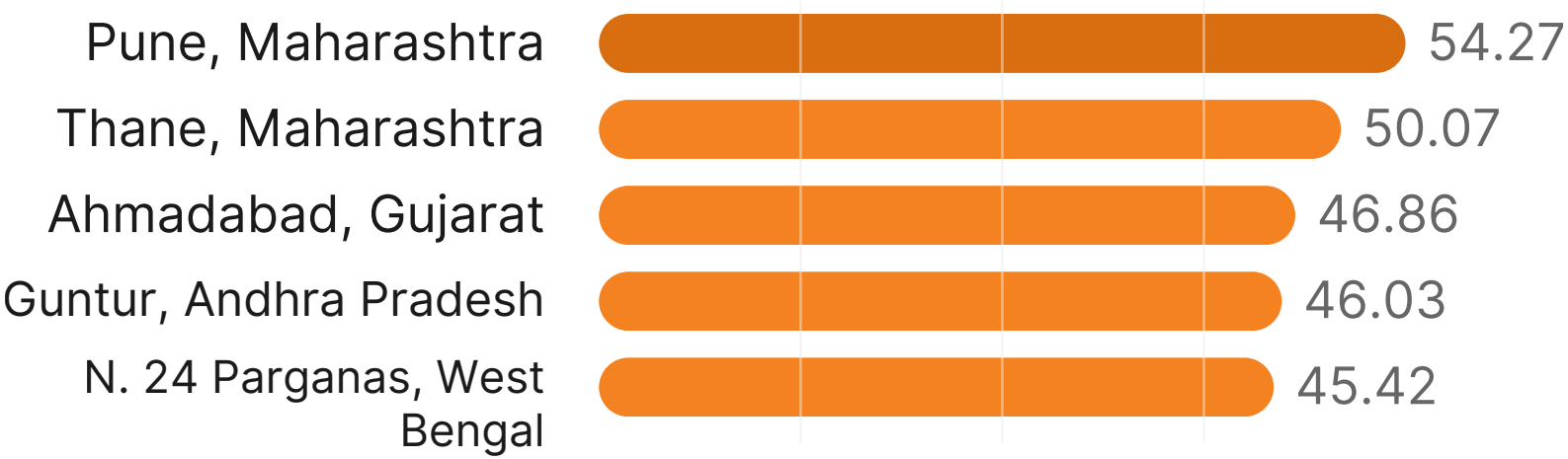
- Pune and Thane lead nationally, reflecting strong market size, healthcare access, and economic capacity.
- Kerala contributes 3 of the Top 10 districts despite relatively smaller populations, driven by excellent healthcare and development indicators.
- Only 0.2% of harmonized State–District records achieve a PMAF score above 50, indicating that high-opportunity markets are highly concentrated.

Chronic vs. Acute Disease Opportunity

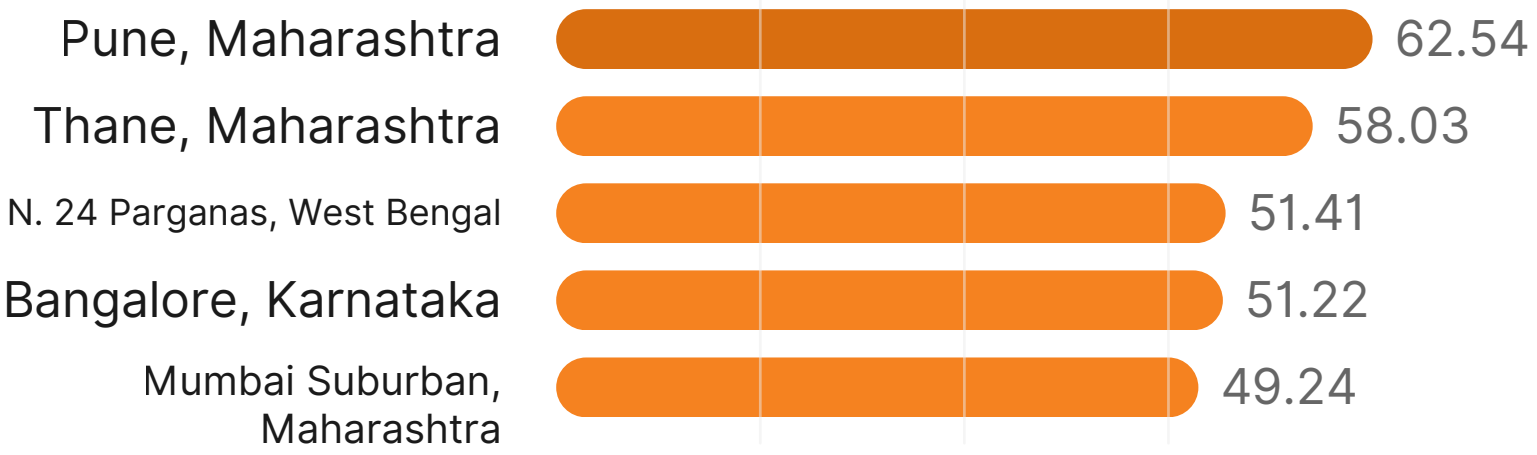
Overall (Top 5)



Chronic PMAF (Top 5)



Acute PMAF (Top 5)



Chronic-leaning: Guntur (AP) ranks 4th in Chronic but outside the Overall Top 10 — an aging-profile, lifestyle-disease market.



Acute-leaning: North 24 Parganas (WB) and Bangalore rank higher on Acute than Chronic — denser child populations and infection burden.



Dual leaders: Pune and Thane top all three indices — the largest, most diversified pharmaceutical markets in the country.

Strategic Business Recommendations



Chronic Focus

Reason: Pune, Thane, Ahmadabad and Guntur post the highest Chronic PMAF scores — driven by anaemia, blood sugar and hypertension prevalence.

Action: Expand specialized chronic-therapy sales reps and staff training in these districts.



Acute Focus

Reason: North 24 Parganas, Bangalore and Mumbai Suburban show elevated Acute PMAF from ARI, diarrhoea and child malnutrition burden.

Action: Prioritize antibiotic and pediatric-therapy outreach; partner with public health campaigns.



Balanced Growth

Reason: Pune and Thane consistently rank among the top performers across Overall, Chronic and Acute PMAF.

Action: Deploy cross-functional teams covering both chronic and acute portfolios.



PMAF Decision Support

Reason: PMAF provides a defensible, data-backed basis for territory and quota decisions, replacing fragmented indicators.

Action: Integrate the dashboard into quarterly sales planning and territory reviews.



Implementation Roadmap & Scalability



Reusable Framework

PMAF code ingests new data (e.g. NFHS-6) with minimal changes to the pipeline.

Custom Modules

New disease vectors can be added by adjusting pillar feature lists.

Enterprise Integration

Dashboard outputs can integrate with enterprise BI/CRM platforms via CSV/API exports.



Innovation & Dashboard Demo

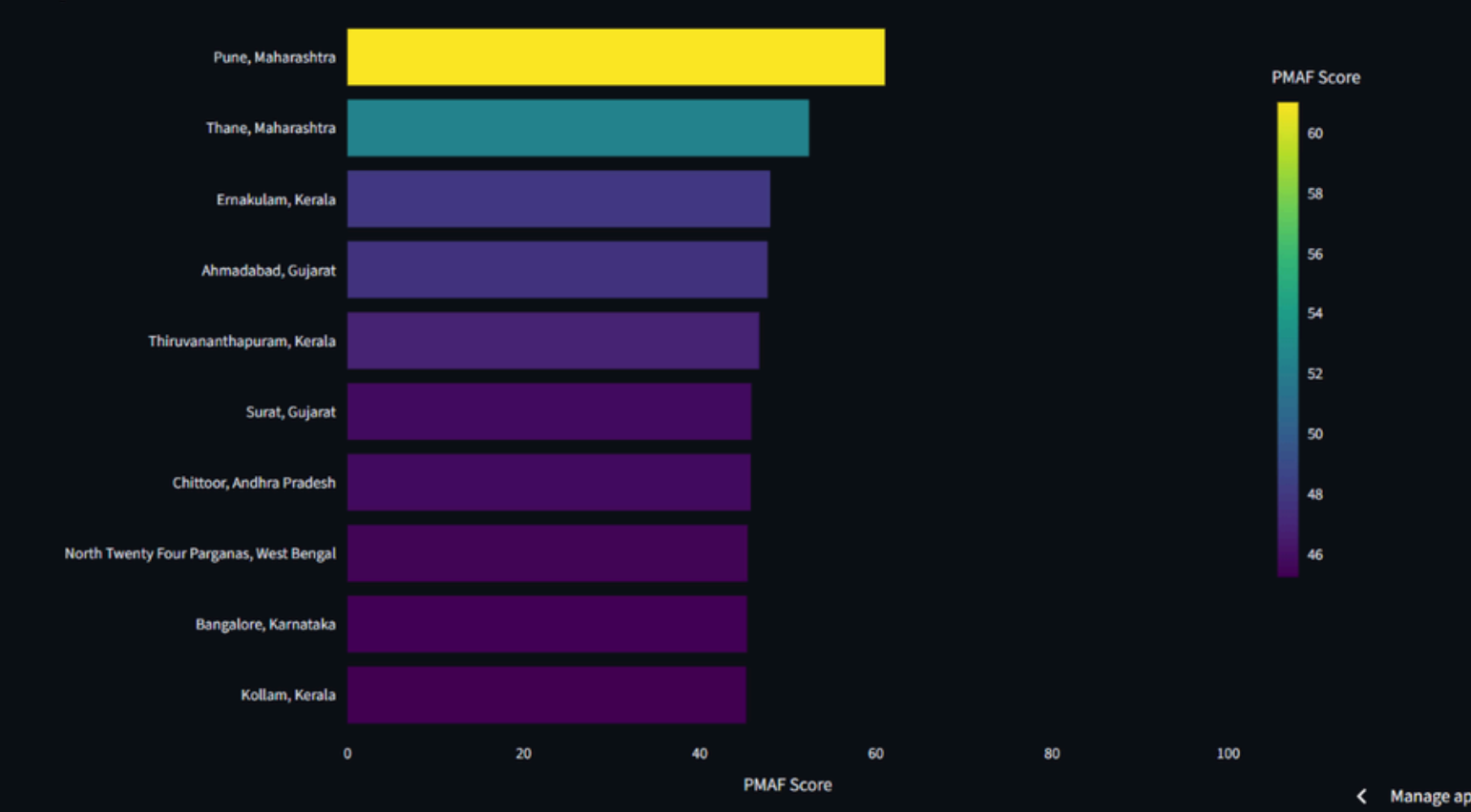
PMAF Dashboard

District Pharmaceutical Market
Attractiveness Framework

Navigation

- ☒ Executive Overview
- ☐ District Explorer
- ☐ Compare Districts
- ☐ Rankings
- ☐ Analytics
- ☐ About PMAF

Top 10 Districts: Overall PMAF



Six Pillars

- > Market Size
- > Chronic Disease
- > Acute Disease
- > Healthcare
- > Economic
- > Development

Data Sources

Census	NFHS-5	RHS	SECC	RBI
Demographic & Economic	Health & Nutrition	Healthcare Infrastructure	Economic & Social	Banking & Finance

Key Features

- Searchable, ranked district table
- Side-by-side Chronic / Acute comparison
- Correlation heatmaps & scatter analytics
- Downloadable CSV reports per ranking

Live PMAF Dashboard

Interactive Streamlit application
(Executive Overview)



Scan to explore the live PMAF Dashboard!

<https://pmaf-trix-sun-pharma.streamlit.app>

Thank You!

Questions?

PMAF transforms public health, healthcare and economic data into actionable pharmaceutical market intelligence, enabling smarter district-level decisions for Sun Pharma.

Team TriX | IIT (ISM) Dhanbad

Live Dashboard • GitHub Repository • PMAF v1.0 | Trilytics 2026

